UNIVERSITY EXAMINATIONS

UNIVERSITEITSEKSAMENS

UNISA university of south africa

MAT1503

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Linear Algebra

Duration 2 Hours

100 Marks

EXAMINERS

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Closed book examination

This paper consists of 4 pages

THE USE OF A POCKET CALCULATOR IS NOT PERMITTED

Answer all the questions

There is a total of 100 marks

[TURN OVER]

QUESTION 1

Let

$$A = \begin{bmatrix} 0 & 0 & 2 & 3 \\ 0 & 1 & -2 & 2 \\ -1 & 2 & 5 & 0 \\ 3 & 1 & 2 & 0 \end{bmatrix}$$

(a) Evaluate the determinant, $\det A$, by expanding along the **last column** (6)

(b) Use (a) above to evaluate the determinant of matrix B, where

$$B = \begin{bmatrix} 3 & 1 & 4 & 3 \\ -1 & 2 & 5 & 0 \\ 0 & 1 & -2 & 2 \\ 3 & 1 & 2 & 0 \end{bmatrix}$$

Hint Note that we obtain the first row of B by adding the fourth row of A to the first row of A

(4)

(c) Prove that the equation of a line through the distinct points (a_1, b_1) and (a_2, b_2) can be written as

$$\begin{vmatrix} x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \end{vmatrix} = 0$$

(6)

(d) Given $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = 3$, evaluate $\det B$, where

$$B = \begin{bmatrix} c+3a & 0 & d+3b \\ 0 & 3 & 0 \\ -2a & 0 & -2b \end{bmatrix}$$

Hint Expand along the 2nd row or 2nd column (5)

(e) Suppose A, B and C are 2×2 matrices such that

$$\det A = 3, \ \det B^{-1} = -2 \text{ and } \det C^T = 4$$

Evaluate

$$\det(ABC)$$ (4)

[25]

[TURN OVER]

QUESTION 2

(a) Find the value(s) of a, b and c such that the linear system

$$\left.\begin{aligned} ax + by - 3z &= -3 \\ -2x - by + cz &= -1 \\ ax + 3y - cz &= -3 \end{aligned}\right\}$$

has the solution $x = 1$, $y = -1$ and $z = 2$ (6)

(b) Consider the linear system

$$\left.\begin{aligned} x - \phantom{y+{}} 3z &= 2 \\ -2x + y + 3z &= 1 \\ 5x - 2y - 9z &= 0 \end{aligned}\right\}$$

(i) Is there a value of a for which $(-1, 2, a)$ is a solution of the above system? If so, give the value of a If not, give reasons why not (3)

(ii) Find all solutions of the above system (6)

(c) Consider the nonhomogeneous system of linear equations

$$\left.\begin{aligned} x + 2y - z &= 4 \\ -y + z &= 3 \\ \left(a^2 + 2a - 3\right) z &= a - 1 \end{aligned}\right\}$$

Determine the value(s) of a for which the system has

(i) no solution, (2)

(ii) exactly one solution, (2)

(iii) infinitely many solutions (2)

(d) Find all values of a for which the associated homogeneous system of (c) has

(i) only the trivial solution, (2)

(ii) nontrivial solutions (2)

[25]

[TURN OVER]

QUESTION 3

(a) Let $\underline{u} = (4, 2, -1)$, $\underline{v} = (3, 1, 1)$ and $\underline{w} = (0, 2, 1)$ Compute the following

- (i) $2\underline{v} - 3\underline{w} - \underline{u}$ (3)
- (ii) $\underline{u} \quad (\underline{w} + \underline{v})$ (3)
- (iii) $\|\underline{u} \times \underline{w}\|$ (4)
- (iv) the orthogonal projection of $\underline{u}$ on $\underline{w}$ (3)
- (v) the vector component of $\underline{u}$ orthogonal to $\underline{w}$ (1)

(b) Let

$$P = (1, 3, 2), \ Q = (2, 3, 1) \quad \text{and} \quad R = (2, 2, 3) \ \text{ be points in } \mathbb{R}^3$$

- (i) Find the area of ΔPQR (4)
- (ii) Find the equation of the plane V passing through P, Q and R (5)
- (iii) Write down the normal of V (2)

[25]

QUESTION 4

(a) Let $z_1 = 1 + \sqrt{2}i$ and $z_3 = 1 - \sqrt{2}i$

- (i) Determine the polar form of z_1 (6)
- (ii) Determine the polar form of z_2 (6)
- (iii) Use the polar forms of z_1 and z_2 to verify that $z_1 \quad z_2 = 3$ (4)
- (iv) Use the polar forms of z_1 and z_2 to verify that $\dfrac{-1}{3} + \dfrac{2}{3}\sqrt{2}i = \dfrac{z_1}{z_2}$ (4)

(b) Use de Moivre's Theorem to derive a formula for the 4th roots of 8 (5)

[25]

TOTAL [100]